

STAT 5014 Homework 1

Your Name

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1 Setup and definitions

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F$ with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$. The sample mean and sample variance are

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \quad \text{and} \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

Definition 1. An estimator $\hat{\theta}$ is *unbiased* for θ if $\mathbb{E}[\hat{\theta}] = \theta$ for every θ in the parameter space.

2 A short derivation

The variance of a random variable X with mean μ can be rewritten as

$$\text{Var}(X) = \mathbb{E}[X^2] - \mu^2. \tag{1}$$

To see this, expand the definition:

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] \tag{2}$$

$$= \mathbb{E}[X^2 - 2\mu X + \mu^2] \tag{3}$$

$$= \mathbb{E}[X^2] - 2\mu \mathbb{E}[X] + \mu^2 \tag{4}$$

$$= \mathbb{E}[X^2] - \mu^2, \tag{5}$$

which is Equation (1).

3 A theorem

Theorem 1 (Central Limit Theorem). *Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F$ with mean μ and finite variance $\sigma^2 > 0$. Then*

$$\sqrt{n} (\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2) \quad \text{as } n \rightarrow \infty.$$

Proof. Omitted; see [Casella and Berger \[2002, Ch. 5\]](#). □

Note that Theorem 1 requires only finite variance, not normality of F .

4 Matrices and piecewise functions

Let $(Y_1, Y_2)^\top$ be bivariate normal with correlation ρ and marginal standard deviations σ_1 and σ_2 . Its covariance matrix is

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \rho \sigma_1 \sigma_2 \\ \rho \sigma_1 \sigma_2 & \sigma_2^2 \end{pmatrix}, \quad \rho \in [-1, 1].$$

The indicator function of the event $\{X > 0\}$ is

$$\mathbf{1}\{X > 0\} = \begin{cases} 1 & \text{if } X > 0, \\ 0 & \text{otherwise.} \end{cases}$$

5 A small simulation study

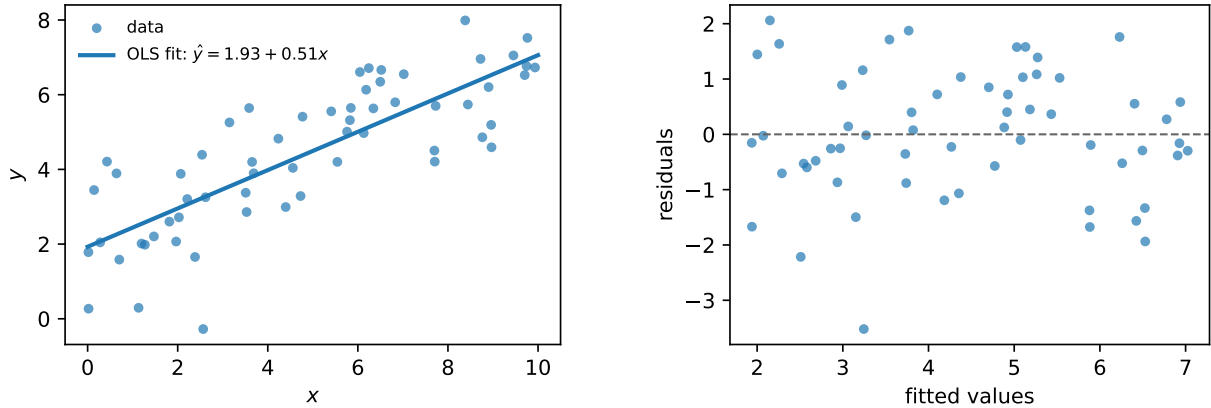
Suppose we simulate data from the model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \quad (6)$$

and estimate β by ordinary least squares, $\hat{\beta} = (X^\top X)^{-1} X^\top y$. The settings are given in Table 1, and the fit for the baseline scenario appears in Figure 1.

Table 1: Settings for the simulation study.

Scenario	Sample size n	Noise SD σ	Replicates
Baseline	60	1.0	1000
Larger	200	1.0	1000
Noisier	60	2.5	1000



(a) Fitted model.

(b) Residual diagnostic.

Figure 1: Simulated data from Equation (6) with an ordinary least-squares fit. Panel b shows no obvious departure from the constant-variance assumption.

Uncertainty in $\hat{\beta}$ could also be assessed without distributional assumptions by resampling [Efron, 1979], and a penalized alternative is available when p is large [Tibshirani, 1996].

References

- George Casella and Roger L. Berger. *Statistical Inference*. Duxbury, Pacific Grove, CA, 2nd edition, 2002.
- Bradley Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1): 1–26, 1979.
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